Unit 2 – Selection and Iteration Name: **KEY**
HW 1

1) Fill in the following blanks with the correct expression using `num` in order to print the appropriate statement.

```
int num = /*value not shown*/;

if(num > 10)
{
     System.out.print("num is greater than 10");
}

if(num <= 10)
{
     System.out.print("num is less than or equal to 10");
}

if(num > 0)
{
     System.out.print("num is positive");
}

if(num < 0)
{
     System.out.print("num is negative");
}

if(num % 2 == 0)
{
     System.out.print("num is even");
}

if(num % 2 == 1)
{
     System.out.print("num is odd");
}

if(num % 5 == 0)
{
     System.out.print("num is evenly divisible by 5");
}

if(num == 0)
{
     System.out.print("num is equal to 0");
}
```

2) Trace through the following code segments. What is the value of `answer` after the following code has executed?

a)

```
int answer = 10;
if(answer > 10) {
     answer++;
}
else if(answer < 20) {
     answer += 2;
}
else if(answer == 10) {
     answer--;
}
```

answer
~~10~~
12

b)

```
double answer = 16.25;
if(Math.sqrt(answer) < 4.5) {
     answer += 0.75;
}
else if(Math.pow(6, 2) >= answer) {
     answer -= 0.25;
}
else if(Math.abs(answer) > 0) {
     answer *= 2;
}
```

answer
~~16.25~~
17.0

c)

```
String answer = "APCSA";
if(answer.substring(0, 2).equals("AP")) {
     answer = answer.substring(2);
}
else if(answer.substring(0, 2).equals("APC")) {
     answer = answer.substring(0);
}
```

answer
~~APCSA~~
CSA

d)

```
boolean boo = true;
int answer = 10;
if(boo) {
     answer++;
}
else {
     answer--;
}
```

answer
~~10~~
11

3) Write a program segment that simulates rolling two dice and then checks to see if the sum is 7 or higher. If it is, print off "The sum is 7 or higher". If the sum is not higher than 7, print off "The sum is not higher than 7".

```
int roll1 = (int)(Math.random()*6) + 1;
int roll2 = (int)(Math.random()*6) + 1;

if(roll1 + roll2 >= 7){
     System.out.println("The sum is 7 or higher");
}
else {
     System.out.println("The sum is not higher than 7");
}
```

C 4) What is the output of the following code?

```
int x = 9;
if(x / 2 <= 4)
{
     x++;
}
if(x / 2 >= 5)
{
     x += 2;
}
System.out.println(x);
```

x
9 → 9/2 is 4
10 → 10/2 is 5
12

(A) 9
(B) 10
(C) 12
(D) 13

A 5) Consider the following code segment.

```
int x = 7;
if(x < 7)
{
     x = 2 * x;
}
if(x % 3 == 1)
{
     x = x + 2;
}
System.out.print(3 * x);
```

x
7 → 7 is not less than 7
→ 7 % 3 is 1
9

print 3 * 9 which is 27

What is printed as a result of executing the code segment?

(A) 27 (B) 9 (C) 14 (D) 21